

■■ Proportional-Self-Aware Logic Flow

1. The Pre-Flight Check (T-25s to T-20s)

The bot activates 25 seconds before the round close. It must clear four hurdles to proceed; otherwise, it logs a "Skip" and waits for the next cycle.

- **Volatility Check:** Compares price at T-25 and T-20. If movement is greater than \$0.50, it aborts to avoid extreme manipulation.
- **Peg Integrity:** Validates that $\text{Price_UP} + \text{Price_DOWN} \geq 0.98$. If lower, the market is flagged as illiquid.
- **The "Dead Side" Filter:** If either side is at \$0.00, the bot skips as the round is effectively settled.
- **The Ceiling Filter:** If the leading side is already greater than \$0.97, the bot skips (insufficient margin for the 2% profit target).

2. The Entry (T-20s)

Once checks pass, the bot initiates the position:

- **Identify Lead:** Determines the winning side based on the higher price.
- **Order Placement:** Places a Single Buy Limit Order at the current market price for the `FIXED_AMOUNT`.

3. The WebSocket "Self-Aware" Event Loop (T-20s to T-5s)

The bot monitors the Polymarket server for real-time "Fill" messages to execute a proportional exit strategy:

- **Instant Reaction:** Upon receiving a message ("Bought X shares at price Y"), the bot reacts immediately.
- **Proportional Sell:** It submits a Sell Limit Order for exactly X shares at $\text{Price} + 0.02$.
Example: If you target 100 shares but only 10 fill initially, the bot lists those 10 for sale at a 2% profit immediately. If another 20 fill a second later, it lists those 20. This ensures maximum exposure to the spread.

4. The Kill Switch (T-5s)

At 5 seconds remaining, the bot cleans up the books:

- **Cancel Buys:** Executes a `cancel_all` command for any unfilled Buy portions to prevent late-entry risk.
- **Maintain Sells:** Leaves all Sell orders active. If they remain unfilled by T-0, they will settle at \$1.00 automatically since the bot holds the winning side.